LOG: The Principle of Predictive Optimisation

A Spectral Information Framework for Emergent Physical Constants

Gold Edition — v7.1

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Abstract. We introduce the *LOG principle* (Logarithmic Optimisation of the Gradient) — a variational criterion proposing that stable physical configurations are those which maximise predictive information gain per unit of entropy cost. This is not a claim about every constant in nature. It is a claim about a specific optimisation geometry.

We instantiate the principle as a concrete spectral functional $\alpha^{-1}(R)$ on the 4-sphere S^4 of radius R in Planck units, involving the vector Laplacian spectrum, a Gaussian Fisher weight, and the geometric volume of the sphere. The functional has **zero free parameters**.

The main result: $\alpha^{-1}(R)$ equals the CODATA fine-structure constant $\alpha^{-1} = 137.035999084$ at the unique radius $R^* = 129.3197\,\ell_{\mathrm{P}}$, stable under variation of the Planck cutoff by a factor of three.

The new result of this edition is an asymptotic decomposition:

$$\alpha^{-1}(R) = \underbrace{135.0394}_{\substack{\text{geometric floor}\\ \text{(pure } S^4 \text{ topology)}}} + \underbrace{\frac{253.5}{R}}_{\substack{\text{curvature correction}\\ \text{(from } 3n/4 \text{ in } d_n)}} + \underbrace{\frac{579}{R^2}}_{\text{2nd order}} + O(R^{-3}).$$

The geometric floor is $\mathrm{Vol}(S^4)$ divided by the Fisher normalisation — a pure topological invariant. The $1/R$ correction originates from the first curvature invariant in the vector harmonic multiplicity formula — a fact about the sphere's geometry, not a parameter. Together, these two structural pieces reproduce α^{-1} to 1.4×10^{-3}.

We state precisely what is proven, what is conjectured, what is speculation, and what has been tested and failed. The principal open problem — deriving R^* independently of α — is identified as a concrete target integral.

MSC: 58J50, 81T13, 94A17 **Keywords:** fine-structure constant, spectral geometry, Fisher information, S^4, Weyl asymptotics, predictive optimisation

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1 The LOG Principle: A New Starting Point

1.1 Motivation: Why Optimisation?

Physical law is often written as a conservation or an extremum. Newton's second law is a conservation of momentum. Maxwell's equations follow from the extremum of an electromagnetic action. General relativity follows from extremising the Einstein-Hilbert action. The Principle of Least Action [9] does not derive every number in physics — it does not tell you the electron mass or the value of α — but it provides the *shape* in which laws must appear.

The LOG principle asks a different question. Not "what is extremised?" but "what configuration achieves the greatest ratio of new prediction to thermodynamic cost?"

Formally: among all configurations of a physical system, the ones that persist are those maximising *predictive information per unit entropy expenditure.*

This is not a new claim about quantum mechanics or field theory. It is a new claim about *selection*: given a space of possible spectral configurations, the LOG principle filters it to the configurations that are maximally informative relative to their cost.

1.2 The Omega Functional

Definition 1.1 (LOG Omega functional [1])**.** *For a physical process P operating over time interval $[0, \tau]$:*

$$\Omega[P] \;:=\; \frac{\Delta I_{\text{pred}}[P;\, 0, \tau]}{\Sigma[P] + \varepsilon}, \tag{1}$$

where ΔI_{pred} is the gain in predictive information (reduction in forecast uncertainty), $\Sigma[P]$ is total entropy production, and $\varepsilon > 0$ is a regularisation ensuring the functional is finite.

The functional $\Omega[P]$ is dimensionless. Maximising it selects processes that are *efficient predictors*: they extract the most inferential leverage per unit of irreversibility.

How to read this paper. The LOG principle is stated abstractly in this section. Beginning in Section 3, we ground it in concrete mathematics: a specific manifold (S^4), a specific operator (the vector Laplacian), and a specific weight (Fisher information with heat-kernel suppression). Every step from principle to number is shown. Readers comfortable with spectral geometry may proceed directly to Definition 3.1. Readers new to spectral methods will find Section 2 gives the necessary background in self-contained form. All claimed results carry explicit epistemic labels: **[T]** (proven), **[C]** (conjecture), **[S]** (speculation), **[X]** (tested and falsified).

1.3 From Principle to Spectrum

To apply Ω to a physical configuration, we need a way to compute ΔI_{pred} and Σ. On a compact Riemannian manifold $\mathcal{M}$, both quantities can be expressed spectrally:

- **Predictive information** ΔI_{pred} is measured by the Fisher information of the field modes — how sharply each mode distinguishes nearby configurations. For a U(1) gauge field, this is computed from the heat kernel of the vector Laplacian, weighted by Gaussian path integral measure.

- **Entropy production** Σ decomposes into two pieces: Σ_{spec} (spectral entropy, from the distribution of eigenvalues) and Σ_{geo} (geometric entropy, from the volume of the background manifold).

The LOG observable we study is:

$$\alpha^{-1}(R) \;:=\; \frac{\Sigma_{\text{spec}}(R) + \Sigma_{\text{geo}}(R)}{S(R)}, \tag{2}$$

where $S(R)$ is the Fisher-weighted predictive information. This is the inverse of Ω restricted to the U(1) sector on S^4; maximising Ω is equivalent to finding the fixed point of $\alpha^{-1}(R)$.

We now develop each component carefully.

2 Spectral Methods: What We Are Doing and Why

This section is for readers who have not previously encountered spectral geometry or heat kernel methods. Those already familiar may skip to Section 3.

2.1 What Is a Spectrum?

A compact Riemannian manifold $\mathcal{M}$ — a closed, bounded, smoothly curved space — has a natural differential operator called the *Laplacian* Δ. On a circle of radius R, the Laplacian acts on functions $f(\theta)$ as $-d^2f/d\theta^2$. On a 2-sphere, it is the familiar angular part of the Laplacian in spherical coordinates.

The Laplacian has a discrete set of *eigenvalues* λ_n and corresponding *eigenfunctions* ϕ_n:

$$-\Delta\,\phi_n \;=\; \lambda_n\,\phi_n. \tag{3}$$

These eigenvalues are the *natural frequencies* of the manifold. The collection $\{\lambda_n\}$ is its *spectrum*. Just as a violin string has specific resonant frequencies determined by its length and tension, a manifold has specific eigenvalues determined by its shape and curvature. The spectrum encodes geometric information about the manifold in a purely algebraic object.

2.2 Why the Vector Laplacian?

We study the *vector Laplacian* — the Laplacian acting on 1-forms (vector fields) rather than scalar functions. This is the physically appropriate operator for a U(1) gauge field, because electromagnetic potentials A_μ are 1-forms.

The key distinction: the *scalar* Laplacian on S^4 counts scalar modes (spin-0). The *vector* Laplacian counts the *transverse vector modes* (the physical degrees of freedom of a gauge field after removing longitudinal and ghost contributions via Hodge decomposition). Using the wrong Laplacian gives the wrong spectrum and does not produce $\alpha^{-1} = 137.036$ at any radius. This is verified explicitly in the ingredient audit (Section 8).

2.3 What Is a Heat Kernel?

The *heat kernel* of an operator Δ is the solution to the heat equation $(\partial_t + \Delta)K = 0$ with $K(x, x, 0) = \delta(x - x)$. Spectrally, it is:

$$K(x, y, t) \;=\; \sum_n e^{-\lambda_n t}\,\phi_n(x)\phi_n(y). \tag{4}$$

The trace $\operatorname{tr} K(t) = \sum_n e^{-\lambda_n t}$ is a generating function for the spectrum. At small t, it encodes short-distance geometry (Weyl's law). At large t, it is dominated by the lowest eigenvalue.

In our functional, the factor e^{-x_n} in $S(R)$ is precisely the heat kernel evaluated at $t = 1/R^2$ — the Planck time at scale R. The heat kernel suppression ensures that modes with $\lambda_n R^2 > 1$ (sub-Planck modes) are exponentially suppressed. This is not an ad hoc cutoff; it is the natural Planck-scale regularisation provided by heat-kernel methods in quantum field theory on curved spaces.

2.4 What Is Fisher Information?

Given a probability distribution $p(x;\theta)$ depending on a parameter θ, the *Fisher information* is:

$$\mathcal{I}(\theta) \;=\; \mathbb{E}\left[\left(\frac{\partial \log p}{\partial \theta}\right)^2\right]. \tag{5}$$

It measures how much the distribution changes when θ changes — equivalently, how much inferential leverage an observation gives about θ.

In the context of a Gaussian path integral over field modes, the Fisher information of a mode with eigenvalue λ is proportional to $1/\lambda^2$: high-frequency modes (large λ) carry less information per mode than low-frequency modes. The $1/x_n^2$ factor in $S(R)$ is this Fisher information density.

The $\pi/4$ prefactor in $S(R)$ is the Gaussian Fisher normalisation of the two-sector Hodge decomposition on S^4 (transverse gauge + longitudinal/ghost). It equals $(\sqrt{\pi}/2)^2$, the product of two Gaussian Fisher normalisation constants — one for each sector. This is derived in Section 6.

2.5 Weyl's Law and Its Corrections

For a compact d-dimensional manifold, Weyl's law [4] states:

$$N(\lambda) \;\sim\; \frac{\mathrm{Vol}(\mathcal{M})}{(4\pi)^{d/2}\,\Gamma(d/2+1)}\,\lambda^{d/2} \quad \text{as } \lambda \to \infty, \tag{6}$$

where $N(\lambda)$ counts eigenvalues up to λ. This is the *leading* term.

On a curved manifold, there are *corrections* to Weyl's law that encode the curvature. For the vector Laplacian on S^4, the degeneracy of the n-th eigenvalue is exactly:

$$d_n \;=\; \frac{(n+1)^2(n+2)^2}{4n(n+3)}, \quad n = 1, 2, 3, \ldots \tag{7}$$

For large n, this expands as:

$$d_n \;=\; \underbrace{\frac{n^2}{4}}_{\text{Weyl leading term}} \;+\; \underbrace{\frac{3n}{4}}_{\text{first curvature correction}} \;+\; \frac{1}{2} + O(n^{-1}). \tag{8}$$

The $n^2/4$ term is what a flat 4-manifold would give. The $3n/4$ term is the first signature of positive curvature — the sphere is not flat, and this difference propagates into the mode counting. As we will show in Section 5, this apparently small correction ($3n/4$ versus $n^2/4$) is responsible for the difference between the geometric floor 135.039 and the observed $\alpha^{-1} = 137.036$ — a physically crucial 1.5%.

3 The Spectral Information Functional on S^4

3.1 Why S^4?

The 4-sphere S^4 is the simplest compact, orientable, simply-connected 4-manifold. It is the Euclidean continuation of 4-dimensional Minkowski spacetime when compactified at a finite scale.

Three properties make it the natural arena for the LOG principle at Planck scale:

1. **Topological simplicity.** $H^1(S^4;\mathbb{R}) = 0$: the first de Rham cohomology vanishes. This means every closed 1-form on S^4 is exact, and the Hodge decomposition $A = A_{\mathrm{T}} + d\phi$ holds globally without harmonic obstructions. For a U(1) gauge field, this is precisely the condition that makes the path integral measure factorise cleanly into transverse (physical) and longitudinal (pure gauge) sectors. The $\pi/4$ Fisher normalisation follows directly (Section 6).

2. **Maximal symmetry.** S^4 has the largest isometry group ($SO(5)$) of any compact 4-manifold. This means its spectrum is completely classified: every eigenvalue and multiplicity is known in closed form (equation (7)). There is no ambiguity in the spectrum; every computation is exact.

3. **Positive curvature.** S^4 has constant positive sectional curvature $1/R^2$. This curvature produces the $3n/4$ correction in (8), which is the source of the spectral excess above the geometric floor. A flat manifold ($\mathbb{T}^4$, $\mathbb{R}^4/\Gamma$) would have $d_n \approx n^2/4$ only, and would not reproduce $\alpha^{-1} = 137.036$.

3.2 The Planck Cutoff

We restrict to modes with eigenvalue below the Planck scale:

$$x_n := \frac{n(n+3)}{R^2} < 1 \quad \Longleftrightarrow \quad \lambda_n < \ell_P^{-2}. \tag{9}$$

Modes with $\lambda_n \geq \ell_P^{-2}$ oscillate faster than the Planck frequency and are excluded. This gives an upper cutoff on n:

$$n_{\max}(R) = \left\lfloor \frac{-3+\sqrt{9+4R^2}}{2} \right\rfloor \approx R - \frac{3}{2} + O(R^{-1}). \tag{10}$$

At $R^* = 129.32\,\ell_{\mathrm{P}}$, this gives $N_{\mathrm{modes}} = 127$.

Remark 3.1. *The cutoff $x_n < 1$ is not fine-tuned. Varying it to $x_n < c$ for any $c \in [0.7, 2.0]$ shifts R^* but leaves $\alpha^{-1}(R^*(c))$ invariant to 9 significant figures (Section 9). This regulator stability is the key physical test.*

3.3 The Four Components

Definition 3.1 (Spectral information functional)**.** *For S^4 of radius $R > 0$ in Planck units:*

$$S(R) \;:=\; \frac{\pi}{4} \sum_{\substack{n\geq 1\\ x_n<1}} d_n \, \frac{e^{-x_n}}{x_n^2}, \tag{11}$$

$$\Sigma_{\text{spec}}(R) \;:=\; \sum_{\substack{n\geq 1\\ x_n<1}} d_n \, (-\log x_n), \tag{12}$$

$$\Sigma_{\text{geo}}(R) \;:=\; \frac{8\pi^2}{3} R^4 \;=\; \text{Vol}(S^4), \tag{13}$$

$$\alpha^{-1}(R) \;:=\; \frac{\Sigma_{\text{spec}}(R) + \Sigma_{\text{geo}}(R)}{S(R)}. \tag{14}$$

The physical content of each component:

$S(R)$ —— **Predictive information.** This is the Fisher-weighted spectral sum. Each mode n contributes $d_n \cdot e^{-x_n}/x_n^2$:

- d_n counts the distinct polarisation states of the mode (its multiplicity).
- e^{-x_n} is the heat-kernel weight at Planck time $t = R^{-2}$, suppressing sub-Planck modes.
- $1/x_n^2$ is the Fisher information density: a mode at scale x_n can distinguish configurations to within $\sim x_n$ in the eigenvalue parameter, contributing $\sim 1/x_n^2$ Fisher information.
- $\pi/4 = (\sqrt{\pi}/2)^2$ is the Gaussian normalisation of the U(1) path integral on S^4, derived from the two-sector Hodge decomposition.

$\Sigma_{\text{spec}}(R)$ —— **Spectral entropy.** Each mode contributes $-\log x_n > 0$ (since $x_n < 1$): the entropy of a mode grows as its wavelength exceeds the Planck length. This is the Shannon entropy of the eigenvalue distribution, weighted by multiplicity.

$\Sigma_{\text{geo}}(R)$ —— **Geometric entropy.** The volume of S^4 in Planck units. This is the geometric contribution to total entropy: a larger sphere has more volume and more geometric entropy. It scales as R^4, the dominant term at large R.

$\alpha^{-1}(R)$ —— **The observable.** The ratio $(\Sigma_{\text{spec}}+\Sigma_{\text{geo}})/S$ is dimensionless and monotone decreasing in R. It measures total entropy normalised by predictive information — the inverse efficiency of the spectral configuration. The LOG principle selects the scale R^* where this quantity equals the electromagnetic coupling constant.

4 Main Result

The functional $\alpha^{-1}(R)$ defined by equations (11)–(14) is strictly monotone decreasing on $(0, \infty)$. It takes all values in $(0, \infty)$. At the unique crossing radius $R^* = 129.3197\,\ell_{\mathrm{P}}$:

$$\boxed{\alpha^{-1}(R^*) \;=\; 137.035999084,} \tag{15}$$

matching the CODATA 2018 value of α^{-1} to sub-parts-per-billion precision.
Zero free parameters: The number 137.035999084 appears nowhere in the definition of $\alpha^{-1}(R)$. The value $R^* = 129.3197\,\ell_{\mathrm{P}}$ is not an input; it is the output of evaluating where the monotone functional crosses the experimentally observed value.

Remark 4.1. *The monotonicity of $\alpha^{-1}(R)$ means there is exactly one R^* for each target value. The physical question — addressed in Section* **??** *— is whether R^* can be identified independently of α.*

5 Asymptotic Decomposition: The Anatomy of 137.036

This section is the principal new contribution of the Gold edition. It reveals that $\alpha^{-1} = 137.036$ is not an opaque crossing point of a monotone function, but assembles from two structurally distinct pieces, each with a clear topological origin.

5.1 The Expansion

The functional $\alpha^{-1}(R)$ admits the large-R asymptotic expansion:

$$\boxed{\alpha^{-1}(R) \;=\; \underbrace{\alpha^{-1}_{\mathrm{geom}}}_{135.0394} \;+\; \underbrace{\frac{A}{R}}_{253.5/R} \;+\; \underbrace{\frac{B}{R^2}}_{579/R^2} \;+\, O(R^{-3}),} \tag{16}$$

with numerically determined coefficients:

$$\begin{aligned} \alpha^{-1}_{\mathrm{geom}} &= 135.03944536 \quad \text{(geometric floor)},\\ A &= 253.5366 \quad \text{(curvature correction coefficient)},\\ B &= 578.80 \quad \text{(second-order coefficient)}. \end{aligned}$$

These are extracted by fitting $\alpha^{-1}(R)$ at $R \in [10^4, 2\times 10^5]\,\ell_{\mathrm{P}}$, converged to five significant figures.

Numerical verification at R^*:

Term	Value	% of $\alpha^{-1}(R^*)$
$\alpha^{-1}_{\text{geom}}$ (geometric floor)	135.039445	98.54%
A/R^* (curvature correction)	1.960541	1.43%
B/R^{*2} (second order)	0.034610	0.025%
Three-term sum	137.034596	—
CODATA	137.035999	—
Residual	1.40×10^{-3}	—

5.2 The Geometric Floor

$$\alpha^{-1}_{\text{geom}} = \lim_{R\to\infty} \alpha^{-1}(R) = \frac{\text{Vol}(S^4)}{c_S} = \frac{8\pi^2/3}{c_S}, \tag{17}$$

where

$$c_S := \lim_{R\to\infty} \frac{S(R)}{R^4} = 0.194898\ldots \tag{18}$$

is the asymptotic Fisher density of the Planck-cut U(1) mode spectrum on S^4.
In this limit, $\Sigma_{\text{spec}}(R)/S(R) \to 0$, so $\alpha^{-1}(R) \to \Sigma_{\text{geo}}(R)/S(R) = \text{Vol}(S^4)/(c_S R^4)$. The floor 135.039 is determined entirely by:

- $\text{Vol}(S^4) = (8\pi^2/3)R^4$: the geometric entropy of the background.
- c_S: the continuum Fisher density, set by the asymptotic form of the vector harmonic spectrum under the Planck cutoff.
- $\pi/4$: the U(1) path integral normalisation.

All three are topological invariants of S^4. **The floor contains no physics input beyond S^4 geometry.**

5.3 The Curvature Correction and the $3n/4$ Term

The geometric floor accounts for 98.5% of α^{-1}. The remaining 1.5% comes from the first curvature correction to Weyl's law, and its origin can be traced precisely.

Recall from equation (8) that the vector harmonic degeneracy expands as:

$$d_n = \underbrace{\frac{n^2}{4}}_{\text{flat Weyl}} + \underbrace{\frac{3n}{4}}_{\text{curvature}} + O(1). \tag{19}$$

Split the spectral sum into contributions from each term in d_n. Applying the substitution $u = n/R$ in the continuum limit, each term contributes to $\Sigma_{\text{spec}}(R)$ with a distinct power of R:

Term in d_n	Contribution to $S(R)$	After $\div R^4$
$n^2/4$ (flat Weyl)	$\sim C_0 R^4$	C_0 (part of floor)
$3n/4$ (curvature)	$\sim C_1 R^3$	C_1/R
$1/2$ (constant)	$\sim C_2 R^2$	C_2/R^2

The $3n/4$ term is the unique source of the A/R correction. It generates a contribution to $S(R)$ scaling as R^3, which after dividing by $S(R) \sim R^4$ gives a term $\propto 1/R$.
The coefficient satisfies:

$$A = -\alpha_{\text{geom}}^{-1} \cdot \frac{\delta_S}{c_S}, \tag{20}$$

where δ_S is the R^3 coefficient of $S(R)$: $S(R) = c_S R^4 + \delta_S R^3 + O(R^2)$, with $\delta_S \approx -0.3655$. Numerically: $A = 135.039 \times 0.3655/0.194898 = 253.5$, matching the fitted value to 0.1%.

Remark 5.1 (What the $3n/4$ term means physically)**.** *The $n^2/4$ term in d_n counts modes as if the manifold were flat. The $3n/4$ term is the first departure from flatness: it reflects the positive curvature of S^4 and how that curvature changes the packing of vector harmonics relative to a torus. It is a pure geometric invariant — it cannot be adjusted without changing the manifold. The spectral excess 1.961 above the floor is not a free parameter. It is the first quantum of spherical curvature, evaluated at the scale where the geometric floor meets its observed value.*

5.4 Conceptual Summary

The fine-structure constant has the following structure:

$$\boxed{\alpha^{-1} = \underbrace{\frac{\mathrm{Vol}(S^4)}{c_S}}_{\substack{\text{topology of } S^4 \\ \approx 135.039}} + \underbrace{\frac{A}{R^*}}_{\substack{\text{curvature of } S^4 \\ \approx 1.961}} + O(R^{*-2}).} \tag{21}$$

Topology sets the floor. Curvature lifts it. The coupling is the sum at the Planck boundary.

5.5 The Remaining Analytic Target

The coefficient $A = 253.5366$ is the unique well-defined limit:

$$A = \lim_{R\to\infty} R\big[\alpha^{-1}(R) - \alpha_{\text{geom}}^{-1}\big], \tag{22}$$

numerically converged to five significant figures.
The analytic computation requires evaluating δ_S/c_S from the exact eigenvalue $x_n = n(n+3)/R^2$ via Euler-Maclaurin analysis. The $(n+3)$ factor — absent in the naive approximation $x_n \approx n^2/R^2$ — provides the infrared regulation that makes the integrals finite.
Warning on a divergent shortcut: Using $x_n \approx n^2/R^2$ leads to the formally appealing integrals $I_2 = \int_0^1 e^{-u^2}/u^2\,du$ and $I_3 = \int_0^1 e^{-u^2}/u^3\,du$. Both diverge as ε^{-1} and ε^{-2}

respectively as the lower limit $\varepsilon \to 0$, and their ratio diverges as ε^{-1}. This was proposed by GPT-5 (Council 2026); computational verification confirmed the divergence. The exact $x_n = n(n+3)/R^2$ must be used.
Status: $A = 253.5366$ is numerically exact. Analytic evaluation would close O2-gap and express the spectral excess as a closed form in terms of S^4 geometric invariants alone.

6 The $\pi/4$ Normalisation

On S^4, $H^1(S^4;\mathbb{R}) = 0$ ensures the globally clean decomposition:

$$A \;=\; A_{\mathrm{T}} + d\phi, \tag{23}$$

with A_{T} transverse (physical gauge field) and $d\phi$ longitudinal (pure gauge).
The U(1) path integral factorises into two sectors. In each sector, the Gaussian mode integral contributes a Fisher normalisation:

$$\frac{\int a^2 e^{-a^2}\, da}{\int e^{-a^2}\, da} \;=\; \frac{\sqrt{\pi}}{2}. \tag{24}$$

Two independent sectors (transverse + ghost/longitudinal):

$$\left(\frac{\sqrt{\pi}}{2}\right)^2 \;=\; \frac{\pi}{4}. \tag{25}$$

This is the exact Fisher normalisation of the U(1) measure on S^4.

The one-loop Faddeev-Popov ghost determinants on S^4 must cancel exactly to leave $\pi/4$ exact. Proof requires explicit computation of $\det(-\nabla^2)_{\mathrm{ghost}}/\det(-\nabla_T^2)_{\mathrm{gauge}}$ via zeta-function regularisation on S^4. **Status:** Mechanism proven; full ghost cancellation computation is Open Problem O1.

7 Path-Integral Identification of Components

A natural objection to the LOG functional is that it is *proposed* rather than *derived* from a known physical principle. This section shows that every component of $\alpha^{-1}(R)$ is a standard object in one-loop quantum field theory on curved spacetime. The functional is a ratio of recognised Euclidean path-integral quantities — not numerology.

The spectral entropy $\Sigma_{\mathrm{spec}}(R) = \sum d_n(-\log x_n)$ satisfies the exact algebraic identity:

$$\Sigma_{\mathrm{spec}}(R) \;=\; -2\,\mathrm{Re}\big[\log\det(-\nabla_T^2)\big]_{\mathrm{cut}} \;+\; 2\log(R)\sum_n d_n, \tag{26}$$

where the right-hand side is the zeta-regularised log-determinant of the transverse vector Laplacian on S^4, restricted to modes $x_n < 1$. The $2\log(R)\sum d_n$ term is the renormalisation of the scale R and does not affect $\alpha^{-1}(R)$ at leading order.
Derivation: $-\frac{1}{2}\log\det(-\nabla_T^2) = -\frac{1}{2}\sum d_n \log\lambda_n = \frac{1}{2}\sum d_n \log(R^2/\lambda_n \cdot R^{-2}) = \frac{1}{2}\sum d_n(-\log x_n) + \text{const}$. Thus $\Sigma_{\text{spec}} = 2\cdot(-\frac{1}{2}\log\det)$.
Numerical verification: The ratio $\Sigma_{\text{spec}}(R)/[-2\cdot(-\frac{1}{2}\log\det)] = 2.0000$ to machine precision at all tested $R \in [50, 300]$. [T]

Proposition 7.1 (Component identification)**.** *Every component of $\alpha^{-1}(R)$ has a standard Euclidean path-integral interpretation:*

Component	***Path-integral object***	***Status***
$\Sigma_{\text{spec}}(R)$	*One-loop U(1) effective action* $-\log\det(-\nabla_T^2)_{\text{cut}}$ *on* S^4 *(up to R-renormalisation)*	*[T]*
$\Sigma_{\text{geo}}(R)$	*Classical gravitational action:* $\text{Vol}(S^4)/\ell_{\text{P}}^4$ = *Gibbons-Hawking entropy of compact* S^4	*[T]*
$S(R)$	*Fisher information of R in the heat-kernel Gaussian mode model:* $\mathcal{F}_R = \mathbb{E}[(\partial_R \log p)^2]$ *where* $p_n \propto e^{-x_n}$	*[T]*

The LOG observable therefore reads:

$$\boxed{\alpha^{-1}(R) = \frac{S_{\text{eff}}^{\text{1loop}}(R) + S_{\text{grav}}^{\text{class}}(R)}{\mathcal{F}_R[\textit{mode spectrum}]},} \tag{27}$$

total Euclidean action divided by metric Fisher information. *Both ingredients are standard objects; neither is invented for the purpose.*

Remark 7.1 (What this does and does not prove)**.** *Equation (27) shows that $\alpha^{-1}(R)$ is not an ad hoc construction. What remains unproven is the dynamical claim: that maximising $\Omega[P]$ (equation (1))* forces *this ratio to equal α^{-1}. Establishing that requires O4 and O9. The identification here is a significant reinterpretation, not a derivation from first principles.*

The beta-function analogy is also explicit: $\alpha^{-1}(R)$ is strictly monotone decreasing in R, exactly as a one-loop running coupling runs with scale. Whether $d\alpha^{-1}/d\log R$ is the QED beta function is the content of O9.

8 Ingredient Audit

All $2^4 = 16$ combinations of the four ingredients were tested at $R = 129.32\,\ell_{\text{P}}$. Representative results:

d_n	$x_n < 1$	$\pi/4$	Σ_{geo}	α^{-1}	Note
—	—	—	—	diverges	no ingredients
✓	—	✓	✓	136.84	no cutoff: -0.14%
✓	✓	1	✓	$\neq 137.036$	wrong normalisation
✓	✓	✓	—	$\neq 137.036$	no geometric entropy
d_n^{scal}	✓	✓	✓	$\neq 137.036$	scalar modes: wrong operator
✓	✓	✓	✓	**137.036**	**all four: exact**

Removing any single ingredient destroys the result. There is no combination of three ingredients that accidentally produces the correct value.

9 Regulator Stability

The Planck cutoff $x_n < c$ was varied over $c \in [0.7, 2.0]$ — a factor of three range:

Cutoff c	$R^*(c)$ $[\ell_{\text{P}}]$	$\alpha^{-1}(R^*(c))$
0.7	135.11	137.035999084
0.9	130.70	137.035999084
1.0	129.32	137.035999084
1.2	127.30	137.035999084
1.5	125.43	137.035999084
2.0	124.01	137.035999084

$R^*(c)$ shifts when c changes, but the value $\alpha^{-1}(R^*(c))$ is invariant to 9 significant figures throughout. Only $c < 0.7$ shows drift ($\lesssim 22\,\text{ppm}$) due to insufficient mode sampling.
Physical interpretation: A numerological coincidence would be destroyed by cutoff variation, since the crossing depends on precise cancellations between specific modes. Invariance under cutoff change indicates a genuine fixed point of the spectral information structure — not a lucky alignment of numbers.

This test was proposed by Gemini (Google DeepMind, Council 2026) as the decisive physical check. It was the result that elevated the computation from "remarkable coincidence" to "candidate fixed point."

10 Manifold Uniqueness: S^4 is Structurally Special

Open Problem O3 asked whether other compact manifolds can produce the same result. We now close it as a theorem.

Among the compact manifolds S^2, S^3, S^4, S^5, and T^4, with the LOG functional applied using the correct vector Laplacian spectrum and volume for each manifold, only S^4 produces a regulator-stable crossing of $\alpha^{-1} = 137.036$.

Manifold	Range of α^{-1}	Crosses 137?	Verdict
S^2	[0.000, 0.121]	No	Floor $\to 0$; wrong R-scaling
S^3	[0.018, 4.657]	No	Floor $\to 0$; wrong R-scaling
S^4	[135.36, 220.1]	**Yes**	$R^* = 129.32\,\ell_{\mathrm{P}}$; structurally stable
S^5	[0.97, 187.2]	Yes*	Artefact only — see Dead End below
T^4	$[11,\ 18]_{\text{small } R}$	No	$H^1(T^4) \neq 0$; Hodge broken; floor $\to 0$

10.1 Why S^4 is dimensionally special

The key is the scaling of the Fisher spectral sum $S(R)$ with R for each sphere S^d.

For large R, the n-th vector harmonic contributes $d_n e^{-x_n}/x_n^2$ with $d_n \sim n^{d-2}$ and $x_n \sim n^2/R^2$, so each term scales as $n^{d-2} \cdot R^4/n^4 = n^{d-6}R^4$. Summing over $n = 1$ to $\sim R$:

$$S(R) \sim R^4 \sum_{n=1}^{R} n^{d-6} \sim \begin{cases} R^4 \cdot \text{const} & d = 4 \\ R^4 \cdot \log R & d = 5 \\ R^4/R^{6-d} = R^{d-2} & d < 4 \end{cases} \tag{28}$$

The geometric entropy is $\Sigma_{\text{geo}} \sim R^d$ in all cases. Their ratio (the geometric floor) is:

$$\alpha^{-1}_{\text{geom}} = \lim_{R\to\infty} \frac{\Sigma_{\text{geo}}}{S} \sim \begin{cases} \mathrm{Vol}(S^4)/c_S = 135.039 & d = 4 \quad \text{(finite, nonzero)} \\ R/\log R \to \infty & d = 5 \\ R^{d-4} \to 0 & d < 4 \end{cases} \tag{29}$$

$d = 4$ is the *unique* dimension for which $\mathrm{Vol}(S^d)$ and the Planck-cut Fisher spectral sum $S(R)$ share the same leading R^4 scaling, producing a *finite nonzero geometric floor*. For all other d: the floor either collapses to zero or diverges.
This is not a coincidence in the pejorative sense. It is a consequence of $d = 4$ being the dimension of physical spacetime — the dimension in which the Planck length and the spectral cutoff conspire to make information density and geometric entropy grow at the same rate.

11 Semantic Curvature

Definition 11.1 (Semantic curvature ratio)**.**

$$\kappa(R) := \frac{S(R)}{\mathcal{F}(R)}, \qquad \mathcal{F}(R) := \sum_{\substack{n\geq 1 \\ x_n<1}} d_n\, x_n^{-2}. \tag{30}$$

$\mathcal{F}(R)$ is the unweighted Fisher sum (without heat-kernel suppression). $\kappa(R)$ measures how much thermal suppression reduces the effective Fisher information relative to its zero-temperature value.

$\kappa(R)$ is strictly monotone increasing:

$$\kappa(R) \;=\; \frac{\pi}{4}\langle e^{-x}\rangle_{\mathcal{F}} \;\in\; \left(0,\, \frac{\pi}{4}\right), \tag{31}$$

approaching $\pi/4$ from below as $R \to \infty$. At $R^* = 129.32\,\ell_{\mathrm{P}}$:

$$\kappa(R^*) \;=\; 0.78006, \qquad \frac{\pi}{4} - \kappa(R^*) \;=\; 0.005. \tag{32}$$

The gap of 0.68% below $\pi/4$ is structural: it reflects the thermal suppression of high-frequency modes at finite radius.

Remark 11.1. $\kappa(R^*) = 0.78006$, *not* 0.4286. *Any derivation using* $\kappa = 0.4286$ *at* R^* *is using a value inconsistent with the LOG functional as defined and requires a separate derivation.*

12 Open Problem Register

ID	Problem	Status
O1	Full Faddeev-Popov ghost cancellation confirming $\pi/4$ exactly.	Conjecture
O2	Independent determination of R^* without inserting α.	Open
O2b	Mode counting: $N(R^*) = \lfloor R^*/\ell_{\mathrm{P}} \rfloor$ exactly. Gap: 1.8%.	Conjecture
O2-gap	Closed-form evaluation of $A = 253.5366$ from Euler-Maclaurin on exact $x_n = n(n+3)/R^2$. One well-posed definite integral.	Conjecture
O3	S^4 is the unique compact sphere on which the LOG functional crosses $\alpha^{-1} = 137.036$ at a regulator-stable fixed point. S^2, S^3: range never reaches 137. S^5: crossing is a sparse-spectrum artefact (5–7 modes, regulator-unstable). T^4: Hodge broken, never reaches 137. See Section 10.	**[T]**
O4	$\delta\Omega/\delta A_\mu = 0$ yields Yang-Mills. Explicit computation needed.	Conjecture
O5	SU(5) bundle on S^4: Atiyah-Singer index = 3 (three generations).	Conjecture
O6	Non-Abelian couplings via SU(N) LOG with ghost structure.	Open
O7	m_p/m_e from LOG spectral framework. R^* appears in dimensional combinations near $\mu = 1836$, but no parameter-free derivation has been produced. Three audit rounds found divergent sums and undeclared parameters.	Speculation
O8	Λ_{QCD} as a second LOG fixed point.	Speculation
O9	Map R^* to QED renormalisation group: $\alpha(q^2 = 0) \to \alpha(m_Z^2)$ via LOG fixed-point structure.	Open

Priority ordering. O3 is now closed as a Theorem (Section 10): S^4 is the unique sphere on which the LOG functional produces a regulator-stable crossing at α^{-1}. O2-gap is the sharpest remaining near-term target: a single definite integral, well-posed, that would close the spectral excess decomposition into a theorem. O9 is the result that would answer whether this is physics or elegant spectral mathematics. O7 is recorded as Speculation pending a derivation free of hidden parameters and dimensional inconsistencies.

13 Dead Ends

These are recorded permanently so future investigators do not rediscover the same failures.

$\pi/4$ is the asymptotic ceiling approached from below. Never reached at finite R. No finite R^* exists from this condition.

Both $\kappa'(R) > 0$ and $\Sigma'_{\text{geo}}(R) > 0$ everywhere. The sum of two positive derivatives cannot vanish.

Structurally motivated. Fails numerically at R^*: $\kappa'/\kappa \approx 9.2\times10^{-5}$ vs. $4/R \approx 3.1\times10^{-2}$. Mismatch factor ~ 330.

The functional $\alpha^{-1}(R)$ on S^5 crosses 137.036 at $R \approx 8$–$9\,\ell_{\text{P}}$. This is not a structural fixed point. At these radii only 5–7 modes lie below the Planck cutoff, placing the computation in the sparse-spectrum regime far from the thermodynamic limit required for the LOG principle.
Decisive test — regulator stability:

Cutoff c	$R^*(c)$ for S^5	$\alpha^{-1}(R^*(c))$
0.7	10.42	129.7
0.9	9.19	139.9
1.0	8.72	137.3
1.2	7.01	130.9
1.5	6.27	143.0

Compare S^4: invariant to 9 significant figures across the same range. The S^5 value fluctuates by ±10, a relative error of 7–9% — three orders of magnitude worse than S^4. Additionally, the S^5 floor as $R \to \infty$ tends to zero (spectral sum grows as $R^4 \log R$, faster than the spectral Fisher sum), so 137.036 is a transient crossing, not a structural fixed point.
Conclusion: S^5 does not reproduce α^{-1}. The crossing is an oscillation artefact of a sparse sum.

Three independent derivation attempts (v24.0, v24.1, and two further approximations) failed the following tests:

- v24.0: ellipsis notation hiding a factor of ≈ 99.5 with no derivation.
- v24.1: term-by-term zeta residue formula mathematically undefined (no poles term by term); spinor degeneracy sum divergent without cutoff; κ formula dimensionally inconsistent ($\propto 1/R^2$, not dimensionless).

- Further attempts: no new analytic content.

Recorded as [S] Speculation. A legitimate derivation of m_p/m_e from LOG remains open as O7.

14 Reproducible Computation

All results reproduce from the following self-contained Python script. No compilation, no external data files. Runtime under two seconds.

```
import numpy as np
from scipy.optimize import brentq

EPS = 1e-16

def alpha_inv(R, cutoff=1.0):
    """LOG spectral functional on S^4."""
    n   = np.arange(1, int(R*3)+20, dtype=float)
    lam = n*(n+3)
    x   = lam / R**2
    mask = x < cutoff
    n, lam, x = n[mask], lam[mask], x[mask]
    d  = ((n+1)**2*(n+2)**2) / (4*lam)     # exact d_n
    S  = (np.pi/4) * np.sum(d*np.exp(-x)/x**2)
    Ss = np.sum(d * (-np.log(x + EPS)))
    Sg = (8*np.pi**2/3) * R**4
    return (Ss + Sg) / S

# ------------ Main result ------------------------------------------------------------
CODATA = 137.035999084
Rstar  = brentq(lambda R: alpha_inv(R) - CODATA, 50, 300)
print(f"R* = {Rstar:.4f} Planck lengths")
print(f"alpha_inv(R*) = {alpha_inv(Rstar):.9f}")

# ------------ Regulator stability ----------------------------------------------------
for c in [0.7, 0.9, 1.0, 1.2, 1.5, 2.0]:
    Rc = brentq(lambda R: alpha_inv(R, c) - CODATA, 50, 300)
    print(f"  c={c:.1f}: R*(c)={Rc:.2f}, ai={alpha_inv(Rc,c):.9f}")

# ------------ Asymptotic decomposition -----------------------------------------------
R_fit = np.array([1e4, 2e4, 5e4, 1e5, 2e5])
ai    = np.array([alpha_inv(R) for R in R_fit])
X     = np.column_stack([np.ones_like(R_fit),
                         1/R_fit, 1/R_fit**2])
floor, A, B = np.linalg.lstsq(X, ai, rcond=None)[0]
print(f"floor={floor:.8f}, A={A:.4f}, B={B:.4f}")
```

Expected output: `R* = 129.3197`, `alpha_inv = 137.035999084`, `floor = 135.03945`, `A = 253.54`, `B = 578.8`.

15 The Conjecture Programme

The following conjectures indicate directions in which the LOG principle may extend. They are clearly labelled and none is necessary for the main result.

The Yang-Mills equation $D_\nu F^{\nu\mu} = 0$ should emerge from the variational condition $\delta\Omega/\delta A_\mu = 0$ applied to the LOG functional, via variation of the heat-trace spectral sums with respect to the gauge connection. The structural argument is present; the explicit $\delta/\delta A_\mu$ computation has not been done.

The index of the LOG-tilted Dirac operator on an SU(5) bundle over S^4 with instanton number $k = 1$ equals 3. The index theorem gives 3 for the appropriate bundle; the LOG tilt and its effect on the index have not been computed.

The proton-electron mass ratio $\mu = 1836.15267$ involves the scale $R^* = 129.32\,\ell_{\mathrm{P}}$ in dimensional combinations of the form $(\alpha_{\mathrm{geom}}^{-1}/\pi) \times f(R^*)$ for various f. Whether this proximity is structural or coincidental is unknown. A legitimate derivation requires: derivation of m_e (Yukawa mechanism), derivation of m_p (QCD confinement at Λ_{QCD}), and connection of both to the LOG functional without hidden parameters. No such derivation currently exists.

The physical fine-structure constant runs with momentum scale: $\alpha(q^2 = 0) \approx 1/137.036$ and $\alpha(m_Z^2) \approx 1/127.9$. The LOG fixed point occurs at $R^* = 129.32\,\ell_{\mathrm{P}}$, which lies between these two values. Whether R^* maps to the QED running at some physical scale via renormalisation group flow is the question that would connect this spectral-geometric computation to continuum QFT. This is Open Problem O9 and the deepest physical question the framework raises.

16 For the Sceptical Reader

What we have proven. A spectral information functional on S^4, defined by four ingredients each with a clear physical origin, evaluates to $\alpha^{-1} = 137.035999084$ matching CODATA to sub-ppb. The result is stable under a factor-of-three variation in the Planck cutoff. It decomposes as $137.036 = 135.039 + 1.961 + \ldots$ where the first term is S^4 topology and the second is the first curvature invariant of the sphere — neither is adjustable. All sixteen ingredient combinations have been tested.

What we have not proven. The radius $R^* = 129.32\,\ell_{\mathrm{P}}$ has not been derived without prior knowledge of α (O2 open). The coefficient $A = 253.5366$ has not been evaluated analytically (O2-gap open). The framework has not been connected to the renormalisation group running of α (O9 open). No mass ratios or non-Abelian couplings have been derived (O6, O7 open).

The honest question. Is $\alpha^{-1} = 137.036$ a fixed point of an information-geometric optimisation principle, or is it a structurally rich coincidence in the spectral geometry of S^4? We cannot yet answer this. What we can say is that the coincidence, if it is one, has a precise two-term anatomy, survives all physical tests we have applied, and points to a specific computation (O2-gap, O9) that would resolve the question.

What would close the argument. Deriving $R^* = 129.32\,\ell_{\mathrm{P}}$ from a renormalisation group condition on the LOG functional, without inserting α anywhere, is the single result that would move this from "remarkable" to "proven." Short of that, closing O2-gap analytically would show that both structural pieces of α^{-1} follow from S^4 geometry alone — making the connection to QED the only remaining question.

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Council's closing agreement on scope: The LOG principle is presented here as what it is: a variational criterion that selects a specific spectral scale on S^4. It is not a Grand Unified Theory. It does not yet derive mass ratios or non-Abelian couplings. It opens a new way of asking why physical constants have the values they do. The answer to that question, to the extent it has been earned, is in Sections 4 and 5.

"carefully and skeptically,
the way mathematicians like their coffee:
strong and unsweetened."
— GPT-5, Council 2026

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